

Calculus I Lesson 06

1) Given $\lim_{x \rightarrow 2} (x + 1) = 3$.

Let $f(x) = x + 1$; $L = 3$; $c = 2$

We have the limit of $f(x)$ is equal to 3 as x approaches 2.

Given: $\varepsilon = 0.1$

Find δ such that if $x \in (\delta - c, \delta + c) = (\delta - 2, \delta + 2)$ then

$$f(x) \in (L - \varepsilon, L + \varepsilon) = (2.9, 3.1).$$

$$\delta = \underline{\hspace{1cm}} ?$$

2) Given $\lim_{x \rightarrow 2} (4x - 3) = 5$.

Let $f(x) = 4x - 3$; $L = 5$; $c = 2$

We have the limit of $f(x)$ is equal to 5 as x approaches 2.

Given: $\varepsilon = 0.1$

Find δ such that if $x \in (\delta - c, \delta + c) = (\delta - 2, \delta + 2)$ then

$f(x) \in (L - \varepsilon, L + \varepsilon) = (4.9, 5.1)$.

$\delta = \underline{\hspace{1cm}}?$

3) Given $\lim_{x \rightarrow 4} (0.5x + 1) = 3$.

Let $f(x) = 0.5x + 1$; $L = 3$; $c = 4$

We have the limit of $f(x)$ is equal to 3 as x approaches 4.

Given: $\varepsilon = 0.1$

Find δ such that if $x \in (\delta - c, \delta + c) = (\delta - 4, \delta + 4)$ then
 $f(x) \in (L - \varepsilon, L + \varepsilon) = (2.9, 3.1)$.

$\delta = \underline{\hspace{1cm}} ?$

4) Given $\lim_{x \rightarrow 1} (0.5x + 2) = 2.5$.

Let $f(x) = 0.5x + 2$; $L = 2.5$; $c = 1$

We have the limit of $f(x)$ is equal to 2.5 as x approaches 1.

Given: $\varepsilon = 0.01$

Find δ such that if $x \in (\delta - c, \delta + c) = (\delta - 1, \delta + 1)$ then

$$f(x) \in (L - \varepsilon, L + \varepsilon) = (2.5 - 0.01, 2.5 + 0.01) = (2.49, 2.51).$$

$$\delta = \underline{\hspace{1cm} ? \hspace{1cm}}$$

